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SUB: DIGITAL ELECTRONICS & C.O

SUBJECT : DIGITAL ELECTRONICS & COMPUTER ORGANIZATION

TOPIC : MEMORY HIERARCHY AND CACHE MEMORY

serial	Topic
1.	INTRODUCTION TO MEMORY
2.	MEMORY HIERARCHY
3.	LEVELS OF MEMORY HIERARCHY
4.	CACHE MEMORY
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1: INTRODUCTION

Memory is an essential part of a computer system that stores data, instructions, and results. Since different types of memory have different speed, capacity, and cost, computers use a memory hierarchy to achieve better performance at a reasonable cost.

Memory hierarchy arranges storage from fastest and smallest to slowest and largest, such as registers, cache memory, main memory (RAM), and secondary storage.

Cache memory is a small, high-speed memory located close to the CPU. It stores frequently used data and instructions so that the processor can access them quickly. Thus, memory hierarchy and cache memory help reduce access time and improve overall computer performance.

2 .MEMORY HIERARCHY

Memory hierarchy is the arrangement of different types of computer memory according to their speed, cost, and storage capacity.

The main purpose of memory hierarchy is to provide the CPU with fast access to frequently needed data while keeping the overall storage cost low.

Order of Memory Hierarchy:

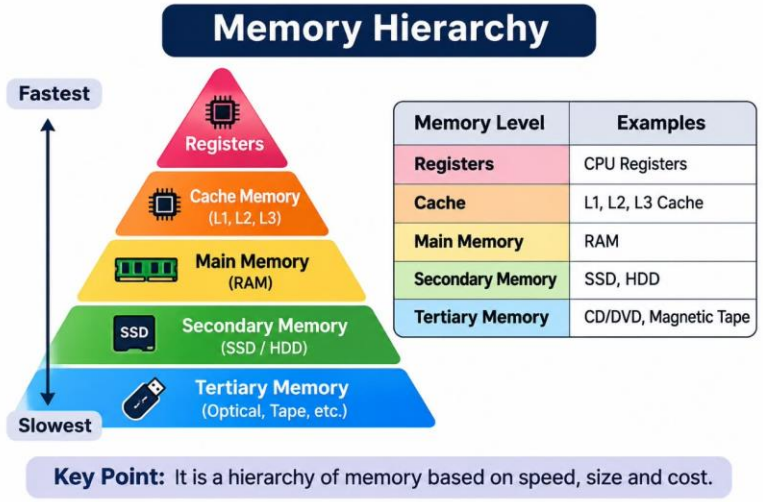
Registers → Cache Memory → Main Memory (RAM) → Secondary Storage (SSD/HDD)

Registers: Fastest and smallest memory.

Cache: Very fast memory used for frequently accessed d

RAM: Larger than cache but slower.

SSD/HDD: Very large storage but slower than RAM.



LEVELS OF MEMORY HIERARCHY

Memory hierarchy is divided into different levels based on speed, capacity, and cost.

Registers

- Located inside the CPU.
- Fastest memory.
- Very small storage capacity.

Cache Memory

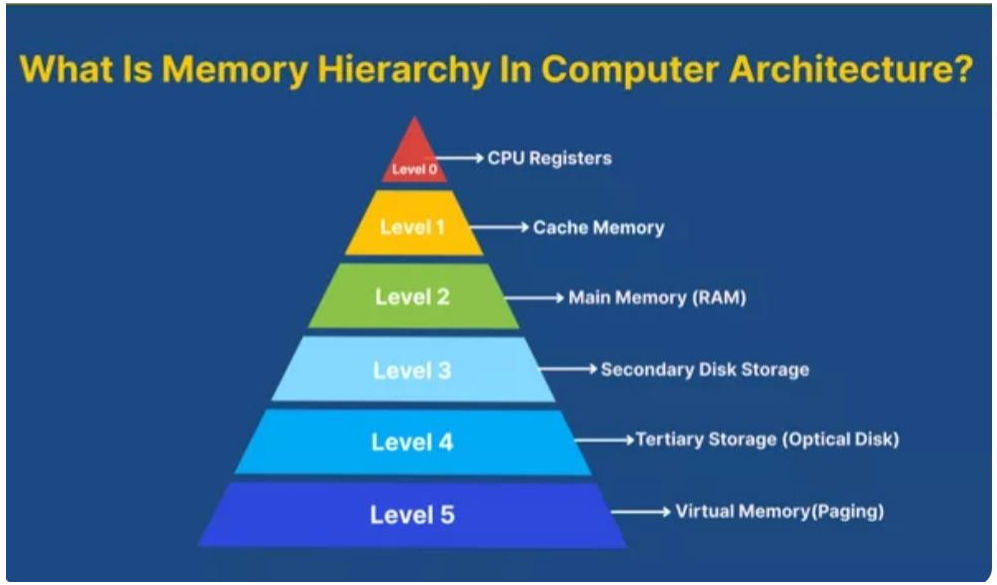
- Located close to the CPU.
- Stores frequently used data and instructions.
- Faster than RAM.

Main Memory (RAM)

- Stores programs and data currently being used.
- Larger than cache but slower.

Secondary Memory

- Includes SSD and HDD.
- Provides large storage capacity.
- Slower than RAM and cache.
- Non-volatile, so data remains after power is turned off.



Memory Hierarchy Order

Registers → Cache → RAM → SSD/HDD

CACHE MEMORY

Cache memory is a small, high-speed memory located inside or very close to the CPU. It stores frequently used data and instructions so that the CPU can access them quickly.

Key Points

It is faster than main memory (RAM).

It is usually made using SRAM (Static RAM).

It reduces the time required by the CPU to access data.

It improves the overall performance of the computer.

It has a smaller capacity than RAM but is more expensive per bit.

Types of Cache Memory

L1 Cache – Smallest and fastest cache.

L2 Cache – Larger than L1 but slightly slower.

L3 Cache – Larger than L2 and usually shared among CPU cores.

In simple words: Cache memory acts as a high-speed bridge between the CPU and RAM, reducing CPU waiting time.

CACHE MEMORY ORGANIZATION

Cache memory organization refers to how cache is structured and how data from main memory is stored and accessed in the cache. The cache is divided into a number of cache lines (blocks), and each line contains data along with a tag used to identify the memory block.

Main Components

Cache Lines/Blocks – Small units that store data copied from main memory.

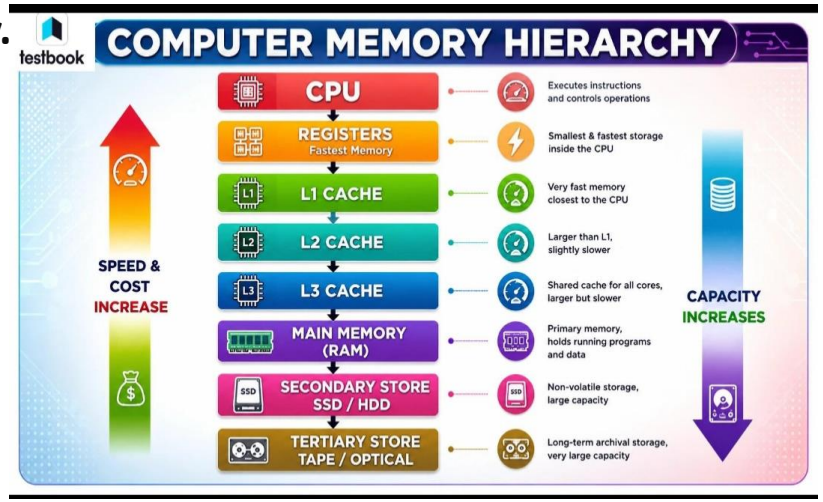
Tag – Identifies which main-memory block is stored in a cache line.

Index – Selects the cache line or set where the data may be located.

Offset – Identifies the exact byte/word within a block.

Valid Bit – Indicates whether the cache entry contains valid data.

Data – The actual information stored in the cache.



Basic Address Format

CPU Address → Tag | Index | Block Offset

Cache Organization Methods

Direct Mapping – Each memory block maps to one specific cache line.

Fully Associative Mapping – A memory block can be placed in any cache line.

Set-Associative Mapping – A memory block maps to a particular set but can occupy any line within that set.

In simple words: Cache organization determines where data is stored in the cache and how the CPU finds it quickly.

CACHE MEMORY MAPPING TECHINIQUE

Cache mapping is the method used to decide where a block of data from main memory will be placed in cache memory. There are three main techniques:

1. Direct Mapping

Each block of main memory is assigned to one specific cache line.

It is simple, fast, and inexpensive.

However, two frequently used blocks may compete for the same cache line.

Formula:
Cache Line = Main Memory Block Number mod Number of Cache Lines

2. Fully Associative Mapping

A memory block can be stored in any cache line.

It provides greater flexibility and reduces conflicts.

However, it requires more complex and expensive hardware.

3. Set-Associative Mapping

Cache is divided into sets, with multiple lines in each set.

A memory block is mapped to a particular set but can occupy any line within that set.

It provides a balance between direct and fully associative mapping.

LOCALITY OF REFERENCE & CACHE PERFORMANCE

Locality of Reference

Locality of reference is the tendency of a program to access the same data or nearby data repeatedly within a short period of time. Cache memory uses this principle to improve CPU performance.

There are two main types:

Temporal Locality

- **If a data item is accessed once, it is likely to be accessed again soon.**
- **Example: A variable used repeatedly inside a loop.**

Spatial Locality

- **If one memory location is accessed, nearby memory locations are likely to be accessed soon.**
- **Example: Accessing elements of an array sequentially.**

Cache Performance

Cache performance shows how effectively the cache reduces memory access time.

CONCLUSION

Memory hierarchy organizes memory according to **speed, capacity, and cost**. Cache memory provides **fast access to frequently used data**, reducing CPU waiting time. Together, they improve the **speed, efficiency, and overall performance** of a computer system.

THANKS YOU..